P_1 补充题6 P_3

$f(x)+xf(-x)=x$

$\downarrow$

$f(-x)-xf(x)=-x$

$\Rightarrow f(x)+x(xf(x)-x)=x$

整理得 $f(x)=\frac{x^2+x}{x^2+1}$

$df(x)=(1+\frac{x}{x^2+1}-\frac{1}{x^2+1})dx$

两边积分得 $f(x)=x+\frac{1}{2}\ln(x^2+1)-\arctan x+C$

P_2 补充题6 P_8

设 l_{MN}: $x=a$.

则 $P_1(a,e^a-1)$ $P_2(a,f(a))$

$|P_1P_2|=e^a-1-f(a)$

$S=\int_0^a f(x)dx=e^a-1-f(a)$

两边求导得 $f(a)=e^a-f'(a)$

$f'(a)+f(a)=e^a$

$f(a)=e^{-\int da}(\int e^a e^{\int da}da+c)$

$=e^{-a}(\int e^{2a}da+c)$

$=\frac{1}{2}e^a+ce^{-a}$

又: $f(0)=0 \to 0=\frac{1}{2}+c \to c=-\frac{1}{2}$

故 $f(x)=\frac{1}{2}(e^x-e^{-x})$.

P_3 补充题6 P_{15} 2).

$y''+y=\sin x\sin 2x$

$\sin x \sin 2x = \frac{1}{2}[\cos(x-2x) - \cos(x+2x)] = \frac{1}{2}(\cos x - \cos 3x)$

$y'' + y = \frac{1}{2}\cos x - \frac{1}{2}\cos 3x$

特征方程: $r^2+1=0 \rightarrow r_1 = i,\ r_2 = -i \rightarrow \cos x,\ \sin x$

对于 $y''+y = \frac{1}{2}\cos x$

设特解 $y_1 = x(A_1\cos x + B_1\sin x)$

$y_1' = A_1\cos x + B_1\sin x + x(-A_1\sin x + B_1\cos x)$

$y_1'' = 2(-A_1\sin x + B_1\cos x) + x(-A_1\cos x - B_1\sin x)$

回代: $-2A_1\sin x + 2B_1\cos x = \frac{1}{2}\cos x \rightarrow \begin{cases} A_1 = 0 \\ B_1 = \frac{1}{4} \end{cases}$

故 $y_1 = \frac{1}{4}x\sin x$

对于 $y''+y = -\frac{1}{2}\cos 3x$

设特解 $y_2 = A_2\cos 3x + B_2\sin 3x$

$y_2' = -3A_2\sin 3x + 3B_2\cos 3x$

$y_2'' = -9A_2\cos 3x - 9B_2\sin 3x$

回代: $-8A_2\cos 3x - 8B_2\sin 3x = -\frac{1}{2}\cos 3x \rightarrow \begin{cases} A_2 = \frac{1}{16} \\ B_2 = 0 \end{cases}$

故 $y_2 = \frac{1}{16}\cos 3x$

通解: $y = \frac{1}{4}x\sin x + \frac{1}{16}\cos 3x + C_1\cos x + C_2\sin x$

P4 补充题16 1)

$y'' + \frac{x}{1-x}y' - \frac{1}{1-x}y = x-1,\ y_1 = e^x$

$y_2 = y_1\int \frac{1}{y_1^2} e^{\int -p(x)dx} dx$

$= e^x \int e^{-2x} e^{\int \frac{x}{x-1}dx} dx$

$= e^x \int e^{-x}(x-1)dx = e^x(-e^{-x}(x-1) + \int e^{-x}dx) = -x.$

齐次方程通解：$y=C_1e^x+C_2x$

设特解 $y^*=C_1(x)e^x+C_2(x)x$

$(y^*)'=C_1'(x)e^x+C_2'(x)x+C_1(x)e^x+C_2(x)$

令 $C_1'(x)e^x+C_2'(x)x=0$ ①

$(y^*)''=C_1'(x)e^x+C_1(x)e^x+C_2'(x)$

回代：$C_1'(x)e^x+C_1(x)e^x+C_2'(x)+\frac{x}{1-x}[C_1(x)e^x+C_2(x)-C_1(x)e^x-C_2(x)x]=x-1$

整理得. $C_1'(x)e^x+C_2'(x)=x-1$ ②

联立 ①,② 得 $\begin{cases}C_1'(x)=xe^{-x}\\C_2'(x)=-1\end{cases}\Rightarrow$ 取 $\begin{cases}C_1(x)=-(x+1)e^{-x}\\C_2(x)=-x\end{cases}$

特解：$y^*=-(x+1)-x^2=-(x^2+x+1)$

通解：$y=-(x^2+x+1)+C_1e^x+C_2x$

$=-(x^2+1)+C_1e^x+C_2x$

P5 补充题6 P20

令 $e^x=t$

则：$\frac{dy}{dx}=\frac{dy}{dt}\frac{dt}{dx}=t\frac{dy}{dt}$

$\frac{d^2y}{dx^2}=\frac{dt}{dx}\frac{dy}{dt}+t\frac{d}{dx}\left(\frac{dy}{dt}\right)=t\frac{dy}{dt}+t\frac{d}{dt}\left(\frac{dy}{dt}\right)\frac{dt}{dx}=t\frac{dy}{dt}+t^2\frac{d^2y}{dt^2}$

$y''-(2e^x+1)y'+e^{2x}y=0$

$t\frac{dy}{dt}+t^2\frac{d^2y}{dt^2}-(2t+1)t\frac{dy}{dt}+t^2y=0$

$\frac{d^2y}{dt^2}-2\frac{dy}{dt}+y=0$

特征方程：$r^2-2r+1=0$ $\rightarrow$ $r=1$ (二重) $\rightarrow e^t, te^t$

通解：$y=e^t(C_1+C_2t)$

回代 $e^x=t$ $\rightarrow$ $y=e^{e^x}(c_1+c_2e^x)$

P_6 习题7 P_{17}

$\vec{OA}$ 与 x,y,z 轴夹角相等 $\rightarrow$ 分量相同

$\vec{OA}=(\frac{\sqrt{3}}{3},\frac{\sqrt{3}}{3},\frac{\sqrt{3}}{3})$ or $(-\frac{\sqrt{3}}{3},-\frac{\sqrt{3}}{3},-\frac{\sqrt{3}}{3})$

$B=(-1-2,\ 2-(-5),\ 1-1)=(-3,7,0)$ $\quad \vec{OB}=(-3,7,0)$

if $\vec{OA}=(\frac{\sqrt{3}}{3},\frac{\sqrt{3}}{3},\frac{\sqrt{3}}{3})$

$\vec{OA}\times\vec{OB}=(-\frac{\sqrt{3}}{3}\times 7,\ \frac{\sqrt{3}}{3}\times(-3),\ \frac{\sqrt{3}}{3}\times 7+\frac{\sqrt{3}}{3}\times 3)=(-\frac{7\sqrt{3}}{3},-\sqrt{3},\frac{10\sqrt{3}}{3})$

if $\vec{OA}=(-\frac{\sqrt{3}}{3},-\frac{\sqrt{3}}{3},-\frac{\sqrt{3}}{3})$

$\vec{OA}\times\vec{OB}=(\frac{7\sqrt{3}}{3},\ \sqrt{3},\ -\frac{10\sqrt{3}}{3})$

综上，$\vec{OA}\times\vec{OB}=(\mp\frac{7\sqrt{3}}{3},\mp\sqrt{3},\pm\frac{10\sqrt{3}}{3})$

P_7 习题7 P_{19}

$[(\vec{a}+\vec{b})\times(\vec{b}+\vec{c})]\cdot(\vec{c}+\vec{a})=(\vec{a}\times\vec{b}+\vec{a}\times\vec{c}+\vec{b}\times\vec{c})\cdot(\vec{c}+\vec{a})$

$=(\vec{a}\times\vec{b})\cdot\vec{c}+(\vec{a}\times\vec{b})\cdot\vec{a}+(\vec{a}\times\vec{c})\cdot\vec{c}+(\vec{a}\times\vec{c})\cdot\vec{a}+(\vec{b}\times\vec{c})\cdot\vec{c}+(\vec{b}\times\vec{c})\cdot\vec{a}$

$=(\vec{a}\times\vec{b})\cdot\vec{c}+(\vec{b}\times\vec{c})\cdot\vec{a}=2+2=4$

P_8 习题7 P_{20}

$$V=|(\vec{a}\times\vec{b})\cdot\vec{c}|=\left|\begin{vmatrix}2&5&0\\0&3&3\\0&2&-5\end{vmatrix}\right|=42$$

$P_{9,10}$ 习题7 P_{24} 1) 3) 5) 7) 8)

1). π: $\frac{x}{a}+\frac{y}{a}+\frac{z}{a}=1$

代入 $(5,-7,4)$ $\rightarrow a=2$

π: $x+y+z-2=0$

3) $\begin{vmatrix} x-2 & y-1 & z+3 \\ 3 & -2 & 7 \\ 0 & -3 & 7 \end{vmatrix}=0$

化简得: π: $7x-21y-9z-20=0$

5) $y=k$

代入 $(3,2,-7)$ $\rightarrow$ π: $y=2$

7) $\vec{a}=(4,-10,-1)$

$\vec{n}'=(3,5,-1)$

$\vec{n}=\vec{a}\times\vec{n}'=(15,1,50)$

π: $15(x-4)+(y-7)+50(z-2)=0$

即 π: $15x+y+50z-167=0$

8). π: $2x+y+2z+k=0$

$A(-\frac{k}{2},0,0)$ $B(0,-k,0)$ $C(0,0,-\frac{k}{2})$

$V=\left|\frac{1}{3}\times(-\frac{k}{2})\times\frac{1}{2}\times(-k)\times(-\frac{k}{2})\right|=\frac{|k|^3}{24}=1$

$K=\pm 2\sqrt[3]{3}$

故 π: $2x+y+2z\pm 2\sqrt[3]{3}=0$

P_{11-12} 习题7 P_{27}

1) $\vec{n}=(3,2,-6)$

$$d=\frac{|Ax_0+By_0+Cz_0+D|}{\sqrt{A^2+B^2+C^2}}=\frac{|3-6-12-1|}{\sqrt{9+4+36}}=\frac{16}{7}$$

2). $\vec{n}=(3,2,-6)$

$$d=\frac{|D_1-D_2|}{\sqrt{A^2+B^2+C^2}}=\frac{21}{7}=3$$

3) $\pi: x+2y-2z+D=0$

$$d=\frac{|D+1|}{\sqrt{1+4+4}}=2$$

$$D=5 \text{ or } -7$$

故π: $x+2y-2z+5=0$ or $x+2y-2z-7=0$.

P_{13} 习题7 P_{28} 3)

$\vec{n_1}=(2,-6,3)$ $\vec{n_2}=(3,-1,-4)$

$$\theta=\arccos\frac{|\vec{n_1}\cdot\vec{n_2}|}{|\vec{n_1}||\vec{n_2}|}$$

$$=\arccos\frac{|6+6-12|}{|\vec{n_1}||\vec{n_2}|}=\frac{\pi}{2}$$

P_{14-15} 习题7 P_{30} 1), 4)

1) $\vec{n}=(4,-2,-2)$ $\vec{s}=(-2,-7,3)$

$$\varphi=\arcsin\frac{|\vec{n}\cdot\vec{s}|}{|\vec{n}||\vec{s}|}=\arcsin\frac{|-8+14-6|}{|\vec{n}||\vec{s}|}=0 \to \text{平行, 不相交}$$

4) $\vec{n}=(2,3,3)$ $\vec{s}=(3,1,2)$

$\vec{n}\not\parallel\vec{s}$, $\vec{n}\cdot\vec{s}\neq 0$ $\to$ 相交

$$L:\begin{cases}x=-2+3t\\ y=2-t\\ z=-1+2t\end{cases}$$

$\to$ 代入 Π: $2(-2+3t)+3(2-t)+3(-1+2t)-8=0$

$9t-9=0$

$t=1$ $\to$ 交点 $(1,1,1)$

P_{16} 习题7 P_{32}

$\vec{s_1}=(-4,1,1)$ $\vec{s_2}=(2,2,-3)$

$P_1(5,1,2)$ $P_2(0,0,8)$ $\overrightarrow{P_1P_2}=(-5,-1,6)$

$$[\overrightarrow{P_1P_2},\vec{s_1},\vec{s_2}]=\begin{vmatrix}-5 & -1 & 6\\ -4 & 1 & 1\\ 2 & 2 & -3\end{vmatrix}=-25\neq 0 \to \text{异面}$$

$\vec{s_1}\times\vec{s_2}=(-5,-10,-10)$

$$d=\frac{|\overrightarrow{P_1P_2}\cdot(\vec{s_1}\times\vec{s_2})|}{|\vec{s_1}\times\vec{s_2}|}=\frac{|[\vec{s_1},\vec{s_2},\overrightarrow{P_1P_2}]|}{|\vec{s_1}\times\vec{s_2}|}=\frac{|[\overrightarrow{P_1P_2},\vec{s_1},\vec{s_2}]|}{|\vec{s_1}\times\vec{s_2}|}=\frac{25}{\sqrt{25+100+100}}=\frac{5}{3}$$

公垂线 $\vec{s}=(1,2,2)$

设交点 $Q_1(5-4t,1+t,2+t)$ $Q_2(2r,2r,8-3r)$

$\overrightarrow{Q_1Q_2}=(2r+4t-5,2r-t-1,-3r-t+6)$

$$\begin{cases}\overrightarrow{Q_1Q_2}\cdot\vec{s_1}=-9s-18t+25=0\\ \overrightarrow{Q_1Q_2}\cdot\vec{s_2}=17s+9t-30=0\end{cases}\to\begin{cases}s=\frac{5}{3}\\ t=\frac{5}{9}\end{cases}$$

L: 一般式 $\begin{cases}10x-7y+2z-16=0\\ y-z+1=0\end{cases}$

P_{17-18} 习题7 P_{35} 2), 6)

2) $\vec{s_1}=(3,-4,6)$ $\vec{s_2}=(1,2,2)$ $P_0=(2,0,5)$

$\vec{n}=\vec{s_1}\times\vec{s_2}=(-20,0,10)\Rightarrow(-2,0,1)$

π: $-2(x-2)+(z-5)=0$

即 π: $-2x+z-1=0$

6). 设平面束方程: $(x+5y+z)+\lambda(x-z+4)=0$

$(1+\lambda)x+5y+(1-\lambda)z+4\lambda=0$

$\vec{n}=(1+\lambda, 5, 1-\lambda)$　　$\vec{n}_0=(1,-4,-8)$

$$\frac{\pi}{4}=\arccos\frac{|\vec{n}\cdot\vec{n}_0|}{|\vec{n}||\vec{n}_0|}$$

$$\frac{\sqrt{2}}{2}=\arccos\frac{|1+\lambda-20-8+8\lambda|}{\sqrt{(1+\lambda)^2+25+(1-\lambda)^2}\cdot\sqrt{1+16+64}}$$

$$\lambda=-\frac{1}{3}$$

$\pi: x+20y+7z-4=0$

同时，$\pi_2: x-z+4=0$ 也与 $x-4y-8z+12=0$ 夹$\frac{\pi}{4}$角

综上，$\pi: x+20y+7z-4=0$ or $x-z+4=0$.

P_{19} 习题7 P_{36}　1)　4)

1)　$\vec{n}_1=(1,1,1)$　$\vec{s}_1=(1,0,0)$

设交点 $P(x_0,1,-1)$

代入 $x+y+z=1$ $\to$ $x_0=1$ $\to P(1,1,-1)$

$\vec{s}=\vec{s}_1\times\vec{n}=(0,-1,1)$

$l: \frac{y-1}{-1}=\frac{z+1}{1}$

4)　设 $l\cap l_1=P_1(t,t,t)$　$l\cap l_2=P_2(s,-2s,1-s)$

$\vec{s}=(t, t-1, t-1)$　$\vec{s}'=(s, -2s-1, -s)$

$\vec{s}\parallel\vec{s}'$ $\to$ $\frac{t}{s}=\frac{t-1}{-2s-1}=\frac{t-1}{-s}$

若 $t\neq 1$ $\to$ $-2s-1=-s$　$s=-1$

代回 $\frac{t}{-1}=\frac{t-1}{1}\to t=\frac{1}{2}$

$P_1(\frac{1}{2},\frac{1}{2},\frac{1}{2})$　$\vec{s}=(1,-1,-1)$

$l: x=\frac{y-1}{-1}=\frac{z-1}{-1}$

若 $t=0$ $\to$ $l_2: \begin{cases}x=s\\y=-2s\\z=1-s\end{cases}$

$$l:\begin{cases} x=0=s \\ y=1-k=-2s \\ z=1-k=1-s \end{cases} \rightarrow s=0\text{矛盾，故 } t\neq -1$$

综上，$l:\ x=\frac{y-1}{-1}=\frac{z-1}{-1}$

思考题

P1.

a)

1). 设 $P(0,0,0)$ $A(1,0,0)$ $B(0,1,0)$ $C(0,0,1)$

则中心点 $O(\frac{1}{2},\frac{1}{2},\frac{1}{2})$

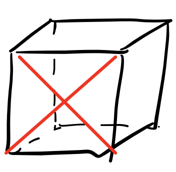
2). π_{ABC}: $x+y+z=1$

$d=\frac{1}{\sqrt{3}}=\frac{\sqrt{3}}{3}$

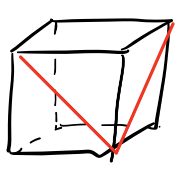
b). 体-体： $\vec{s_1}=(1,1,1)$ $\vec{s_2}=(-1,1,1)$

$\theta=\arccos\frac{-1+1+1}{3}=\arccos\frac{1}{3}$

面-面

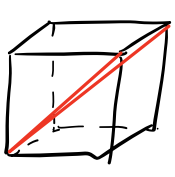


$\theta=\frac{\pi}{2}$

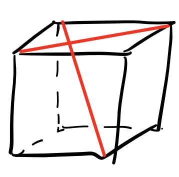


$\theta=\frac{\pi}{3}$

体-面

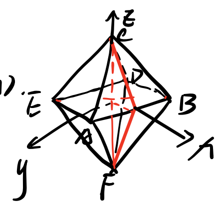


$\theta=\arccos\frac{1+1}{\sqrt{2}\times\sqrt{3}}=\arccos\frac{\sqrt{6}}{3}$



$\theta=\arccos\frac{-1+1}{\sqrt{2}\times\sqrt{3}}=\frac{\pi}{2}$

P2

a) 

$\theta = 2\arccos\frac{\sqrt{3}}{3}$

b). $A(\frac{1}{2},\frac{1}{2},0)$ $B(-\frac{1}{2},\frac{1}{2},0)$ $C(0,0,\frac{\sqrt{2}}{2})$ $\rightarrow \vec{n}_1=(0,\sqrt{2},1)$

$D(-\frac{1}{2},-\frac{1}{2},0)$ $E(-\frac{1}{2},\frac{1}{2},0)$ $F(0,0,-\frac{\sqrt{2}}{2})$ $\rightarrow \vec{n}_2=(0,\sqrt{2},1)$

$\vec{n}_1=\vec{n}_2 \rightarrow$ 面ABC // 面DEF

同理其它面成对平行

$\Pi_{ABC}: \sqrt{2}y+z-\frac{\sqrt{2}}{2}=0$

$\Pi_{DEF}: \sqrt{2}y+z+\frac{\sqrt{2}}{2}=0$

$\rightarrow d=\frac{\sqrt{2}}{\sqrt{2+1}}=\frac{\sqrt{6}}{3}$

P3.

a) $\theta = 2\arccos\frac{\sqrt{6}}{3}$

b) $d=\sqrt{\frac{3}{4}-\frac{1}{12}}=\frac{\sqrt{2}}{3}$

P4

a) 正十二面体: $\varphi=\frac{\sqrt{5}+1}{2}$ 边长先扩大为$\sqrt{5}-1$

$A(\varphi,\frac{1}{\varphi},0)$ $B(\varphi,-\frac{1}{\varphi},0)$ $C(1,1,1)$

$D(1,-1,1)$ $E(1,1,-1)$

$\theta=\arccos\frac{-1}{\sqrt{5}}=\arccos(-\frac{\sqrt{5}}{5})$

正二十面体

$A(1,0,\varphi)$ $B(-1,0,\varphi)$ $C(0,\varphi,1)$ $\rightarrow \vec{n}=(0,\sqrt{5}-1,2+2\sqrt{5})$

$\vec{m}=(1,1,1)$ $\theta=\arccos(-\frac{\sqrt{5}}{3})$

b) 正十二面体

$A'(-\varphi, -\frac{1}{\varphi}, 0)$ $B'(-\varphi, \frac{1}{\varphi}, 0)$ $C'(-1, -1, -1)$ $\rightarrow \vec{n}'$

A', B', C' 分别与 A, B, C 关于原点对称 $\rightarrow$ 法向量共线 $\rightarrow$ 面 $ABC \parallel$ 面 $A'B'C'$ ，面成对平行.

$$d = \frac{2+\sqrt{5}}{\sqrt{3}}$$

正二十面体同理，面成对平行

$$d = \frac{3+\sqrt{5}}{\sqrt{3}}$$